Some more physics-related things to try to plug in

The text marked as red merely serves as a reminder that most of the features cannot be plugged in directly and probably require some ingenuity to implement into the model; in most cases one will unlikely find a straightforward way of accounting for the discussed effects in a careful, *physically-explicit* manner.

Wake effect

It has been discussed in the previous file; the coordiantes are now available in two forms in the same directory this document sits in, but I doubt they will find any use at this stage.

Rolling time windows; turbulent flows

Introducing rolling time window features might help inject "atmospheric memory"into the model
To separate turbulent atmospheric regimes from more calm flows, the model might need to look at the history leading up to the current observation:

- **Steady Flow:** Characterized by low temporal variability. The wind speed and direction over the past few hours remain close to the current values.
- **Turbulent Flow:** Characterized by high temporal fluctuations. The current speed of 10 m/s might be a sudden gust following a calm period, and the wind direction may change rapidly.

Ideas pile

1. **Rolling Standard Deviation ($\sigma_{rolling}$):** This is the primary "statistical indicator" of disorder. A low standard deviation over a 3-hour window signals a laminar, stable flow. A high standard deviation explicitly flags turbulence.
2. **Deviation from the Rolling Mean (Δv):** This feature quantifies how anomalous the current hour is compared to the recent trend:
$$\Delta v = v_{\text{current}} - \mu_{\text{rolling}}$$
If $\Delta v \approx 0$, the flow is steady. If $\Delta v \gg 0$, it indicates a sudden, sharp gust.
3. **Wind Direction Vector Decomposition (U and V Components):** Wind direction $(0° - 360°)$ cannot be averaged directly due to the modular discontinuity at the $360° \to 0°$ boundary. Decomposing the angle θ into Cartesian coordinates might help:
$$U = \cos(\theta), \quad V = \sin(\theta)$$
Calculate the rolling means (μ_U, μ_V) and rolling standard deviations (σ_U, σ_V) on these components. High values of σ_U or σ_V indicate that the wind direction is chaotic.

Alternative: Exponentially Weighted Moving Averages (EWMA)

Vane chatter effect

Mechanical wind vanes often suffer from an artifact known as vane chatter. When exposed to highly turbulent, high-frequency wind shifts, the physical tail of the sensor overcorrects and swings rapidly back and forth.
This creates a high-frequency noise or "chatter"in the raw direction data, which does not represent the true macro-meteorological wind flow.

- **True Directional Shift:** Accompanied by a physical atmospheric driver, such as a drop in temperature, a change in pressure, or a specific wind speed profile. The direction changes and stabilizes at a new angle.
- **Vane Chatter Artifact:** Characterized by extremely high directional variance specifically during low-to-moderate wind speeds (where the vane lacks aerodynamic damping) or extreme gusty conditions. The values oscillate rapidly around a central mean without establishing a new sustained direction.

Another pile of ideas

1. **First-Order Differencing of Vector Components:** Calculate the absolute hour-to-hour change in the U $(\cos\theta)$ and V $(\sin\theta)$ components:
$$\Delta U_t = |U_t - U_{t-1}|, \quad \Delta V_t = |V_t - V_{t-1}|$$
Persistent, large values of ΔU and ΔV from one hour to the next signify high-frequency chatter, whereas a true shift shows a one-time spike followed by stabilization.

2. **Speed-to-Variance Ratio:** Vane chatter is highly dependent on wind speed. At very low speeds, the vane acts like an under-damped oscillation.

$$\text{Chatter\_Index} = \frac{\sigma_{V\_rolling}}{v_{\text{current}} + \epsilon}$$

Where $\sigma_{V\_rolling}$ is the rolling standard deviation of the directional vector, v_{current} is the wind speed, and ϵ is a small constant to prevent division by zero. A high index points directly to low-speed sensor drift/chatter.

However, I highly doubt that all of this is noticeable if our data points are spaced 1 hour apart from each other.

The Air Density – Humidity Relationship

According to Avogadro's Law, equal volumes of gases at the same temperature and pressure contain an equal number of molecules. Dry air is composed primarily of Nitrogen (N_2, molar mass ≈ 28 g/mol) and Oxygen (O_2, molar mass ≈ 32 g/mol), yielding an average molar mass of roughly 28.97 g/mol. When humidity increases, water vapor molecules (H_2O, molar mass ≈ 18.02 g/mol) displace these heavier nitrogen and oxygen molecules. Replacing heavier molecules with lighter ones reduces the total mass per unit volume, decreasing the overall density (ρ) of the air mass.
To calculate the density of moist air (ρ_{moist}), the total atmospheric pressure (P) must be separated into the partial pressure of dry air (P_d) and the partial pressure of water vapor (e) using Dalton's Law:

$$P = P_d + e$$

The density of the moist air mixture is derived from the ideal gas law:

$$\rho_{\text{moist}} = \frac{P_d}{R_d T_K} + \frac{e}{R_v T_K} = \frac{P - e}{R_d T_K} + \frac{e}{R_v T_K}$$

Where T_K is the absolute temperature, R_d is the specific gas constant for dry air (287.05 J/(kg · K)), and R_v is the specific gas constant for water vapor (461.495 J/(kg · K)).
To determine the actual vapor pressure (e), we use the ambient temperature (T in °C) and Relative Humidity (RH as a fraction between 0 and 1) coupled with the **Magnus-Tetens equation** to compute the saturation vapor pressure (e_{sat}):

1. **Calculate Saturation Vapor Pressure (e_{sat}):** The Magnus-Tetens equation provides an empirical, highly accurate approximation for e_{sat} (in hPa or kPa depending on the constants used) within standard meteorological temperature ranges:

$$e_{\text{sat}}(T) = 6.112 \times \exp\left(\frac{17.67 \cdot T}{T + 243.5}\right) \quad [\text{hPa}]$$

2. **Calculate Actual Vapor Pressure (e):** Using the measured Relative Humidity, the actual partial pressure of the water vapor is:

$$e = RH \times e_{\text{sat}}(T)$$

3. **Final Density Equation Integration:** Substituting e into the moist air density formula yields the complete expression:

$$\rho_{\text{moist}} = \frac{P - \left[RH \cdot 6.112 \times \exp\left(\frac{17.67 \cdot T}{T+243.5}\right)\right]}{R_d T_K} + \frac{RH \cdot 6.112 \times \exp\left(\frac{17.67 \cdot T}{T+243.5}\right)}{R_v T_K}$$

Implementation ideas:
Air density scales the kinetic energy of the wind ($E_k = \frac{1}{2}\rho v^2$), as mentioned before. By feeding the explicitly calculated ρ_{moist} directly into the model we might decrease the error. The bad thing is that we do not posess explicit humidity data, once again, possibly rendering this approach somewhat irrelevant.

The Impact of Temperature on Transformer Efficiency and Power Output

Your assumption is physically correct: as the operating temperature of a transformer rises, its **energy conversion efficiency coefficient (η) decreases.
On a hot day, high ambient temperatures reduce the transformer's thermal gradient, impeding its ability to dissipate internally generated heat. This elevates the internal winding temperature, triggering a negative feedback loop that increases internal losses and curtails the maximum safe power output available to the grid.
The decrease in the conversion coefficient η under high temperatures is primarily driven by two types of internal losses:

- **Elevated Winding Losses (Load Losses / I^2R Losses):** The electrical resistance (R) of the copper or aluminum windings increases linearly with temperature according to the relation $R(T) = R_0[1 + \alpha(T - T_0)]$, where α is the positive temperature coefficient of resistance. As R rises on a hot day, more electrical energy is converted into waste heat rather than being transferred to the output.
- **Increased Stray and Eddy Current Losses:** Higher temperatures can slightly alter the magnetic properties of the core steel and structural components, increasing stray electromagnetic losses under high load conditions.

The total useful power output delivered by the transformer (P_{out}) is directly constrained by its temperature-dependent efficiency coefficient $\eta(T)$:

$$P_{\text{out}}(T) = P_{\text{in}} \times \eta(T)$$

Where P_{in} is the power injected into the primary winding. The efficiency coefficient is calculated as:

$$\eta(T) = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{core}} + P_{\text{winding}}(T)}$$

On a hot day, $P_{\text{winding}}(T)$ increases significantly, causing $\eta(T)$ to drop.
Furthermore, transformers are thermally constrained by a maximum allowable hotspot temperature (T_{hotspot}) to protect the solid insulation from accelerated thermal degradation. The hotspot temperature is governed by:

$$T_{\text{hotspot}} = T_{\text{ambient}} + \Delta T_{\text{top-oil}} + \Delta T_{\text{winding-to-oil}}$$

When T_{ambient} spikes on a hot day, T_{hotspot} quickly approaches its critical threshold. To prevent catastrophic insulation failure, something has to derate the transformer. This forces a reduction in the apparent power (S, in MVA) and active power (P, in MW) flowing through the unit, severely restricting total power output precisely when grid demand is often at its peak.
Implementation ideas. We seem to know very little details about the concrete hardware characteristics; as such this section also might be lacking in usefullness.

Approximating the Reynolds Number Without Blade Geometry

The Reynolds number (Re) is a dimensionless parameter that quantifies the ratio of inertial forces to viscous forces within a fluid flow. In wind energy applications, Re heavily influences the lift and drag coefficients (C_L and C_D) of the blades.
When blade geometry—such as the specific airfoil profile, chord length (c), and twist angle—is entirely unknown, we obviously cannot calculate the exact local Reynolds number along the span. However, we can construct a **Macroscopic Airfoil Reynolds Scaling (Re_{macro})** using known environmental data and fundamental wind turbine scaling laws.
The standard equation for the Reynolds number of an airfoil section is expressed as:

$$Re = \frac{\rho \cdot W \cdot c}{\mu}$$

Where ρ is the air density, W is the relative wind velocity seen by the blade segment, c is the chord length, and μ is the dynamic viscosity of air.
Ideas To implement this into our model without knowing c or the exact W, we can try to substitute the unknown terms with macroscopic proxies:
The relative wind speed hitting a blade segment is a vector combination of the ambient wind speed (v) and the rotational blade speed ($u = \Omega \cdot r$, where Ω is the rotor speed and r is the radial position). For most wind turbines operating under normal conditions, the tip speed is much larger than the wind speed ($u \gg v$). Therefore, the relative velocity scales almost linearly with the tip speed:

$$W \propto \Omega \cdot R_{\text{rotor}}$$

The variation in velocity is heavily dominated by the **Rotor Rotational Speed (Ω, in RPM)**. But once again we seem to be completely in the dark regarding the turbine RPM.

The Composite Macroscopic Re Formula

Combining the above approximations yields a dimensionless proxy feature (Re_{proxy}) that we can utilize in the following way:

$$Re_{\text{proxy}} = \frac{\rho_{\text{moist}} \cdot \Omega \cdot v}{\mu(T)} \quad \text{or} \quad Re_{\text{proxy\_simplified}} = \frac{\rho_{\text{moist}} \cdot \Omega}{\mu(T)}$$

Where ρ_{moist} is the calculated air density (accountiong for humidity), Ω is the measured turbine rotor speed (RPM) maybe we can somehow account for not knowing this metric explicitly., v is the ambient wind speed, and $\mu(T)$ is the temperature-adjusted dynamic viscosity.
Aerodynamic performance degrades or changes sharply at certain Reynolds number thresholds due to phenomena like laminar separation bubbles or early stall.
By feeding Re_{proxy} into the model, it enables us to split the data at critical aerodynamic transition zones, even without explicit knowledge about blade geometry:

$$\texttt{if}\ \ Re_{\text{proxy}} < \text{threshold}_{\text{stall}}\ ,\ \Delta v > 0 \implies \text{“Lower aerodynamic efficiency” penalty}$$

This allows the model to capture changes in the lift-to-drag ratio across different weather seasons (winter vs. summer), which directly alters total power production.

Bulk Richardson Number (Ri_b)

The Bulk Richardson Number (Ri_b) is a dimensionless parameter used to assess dynamic stability and the presence of turbulence in the atmospheric boundary layer. It quantifies the ratio of thermally produced buoyancy forces to mechanically produced vertical wind shear forces.
By incorporating Ri_b, we'll be able to provide our model with a macro-scale meteorological indicator. This indicator tells the model whether the atmosphere at the measuring point is stably stratified (suppressing vertical motion and turbulence) or unstable (convectively driven, causing high turbulence and vertical mixing).
The standard definition of the Bulk Richardson Number over a vertical layer of thickness Δz is:

$$Ri_b = \frac{g}{\theta_v} \frac{\Delta\theta_v \cdot \Delta z}{(\Delta u)^2 + (\Delta v)^2}$$

Where g is 9.81 m/s^2, θ_v is the mean virtual potential temperature, $\Delta\theta_v$ is the vertical gradient of virtual potential temperature across the layer, and $\Delta u, \Delta v$ represent the vertical wind shear (change in horizontal wind components over Δz).
We can approximate the thermal and mechanical gradients using **temporal variations as proxies for spatial variations**, supported by Taylor's Frozen Turbulence Hypothesis:

1. **Approximating Thermal Buoyancy ($\Delta\theta_v$):** Instead of a vertical spatial gradient, we can calculate a temporal thermal gradient ($\Delta T_{\text{temporal}}$) using rolling time windows (e.g., the difference between the current temperature and the 3-hour or 6-hour rolling mean).

2. **Approximating Vertical Wind Shear ($(\Delta u)^2 + (\Delta v)^2$):** In our case, high vertical wind shear might correlate strongly with rapid temporal changes in the wind vector as wind shear layers pass through the sensor. Approximate the denominator using the rolling standard deviations or first-order differences of decomposed vector components ($\sigma_U^2 + \sigma_V^2$) over short windows (e.g., 1 to 3 hours)

The Single-Point Proxy Feature ($Ri_{b,\text{proxy}}$)

$$Ri_{b,\text{proxy}} = \frac{g \cdot \Delta T_{\text{temporal}} \cdot z_{\text{sensor}}}{T_K \cdot [(\Delta U_{\text{temporal}})^2 + (\Delta V_{\text{temporal}})^2 + \epsilon]}$$

Where T_K is the current absolute temperature, z_{sensor} is the fixed height of the sensor (acting as a constant multiplier), and ϵ is a small constant to prevent division by zero in case of a flow close to laminar.
The atmosphere typically transitions through three critical regimes based on the Richardson number:

- **Unstable ($Ri_b < 0$):** Convective turbulence. Wind speeds can be highly erratic with strong vertical updrafts and downdrafts.

- **Neutral ($0 \leq Ri_b < Ri_{\text{crit}}$):** Mechanical shear drives the flow; turbulence is purely dynamic (forced convection).

- **Stable ($Ri_b > Ri_{\text{crit}}$, where $Ri_{\text{crit}} \approx 0.25$):** Buoyancy suppresses turbulence, leading to highly predictable, laminar, stratified layers.

By explicitly mapping $Ri_{b,\text{proxy}}$ as an input column, we segment our 3-year dataset into distinct atmospheric stability regimes:

$$\texttt{if}\ \ Ri_{b,\text{proxy}} > 0.25 \implies \text{low-variance steady flow profiles}$$

$$\texttt{if}\ \ Ri_{b,\text{proxy}} < 0 \implies \text{possible chaotic power fluctuations due to convective mixing}$$

This targeted segmentation is aimed at reducing the model's residual errors during seasonal transitions and diurnal cycles (daytime heating vs. nighttime cooling).